

Machine Learning Basics

Supervised Learning

In supervised learning, each training example contains an input and a target output. Classification predicts a category, while regression predicts a numerical value. Examples include spam detection and house-price prediction.

Unsupervised Learning

Unsupervised learning works without labeled target values. The algorithm looks for structure in the data. Clustering groups similar examples, while dimensionality reduction creates a compact representation of high-dimensional data.

Training and Testing

A dataset is commonly divided into training and test portions. The training data is used to fit the model. The test data is held back to estimate how the model performs on unseen examples. A validation set or cross-validation can be used during model selection and hyperparameter tuning.

Overfitting and Underfitting

Overfitting occurs when a model learns training details too closely and performs worse on unseen data. Underfitting occurs when the model is too simple to capture useful patterns. Regularization, appropriate model complexity, more data, and careful evaluation can help control these problems.

Parameters and Hyperparameters

Parameters are learned from the training data, such as weights in a linear model. Hyperparameters are configuration choices set before or around training, such as a learning rate, tree depth, or regularization strength.

Evaluation Metrics

Classification can be evaluated with accuracy, precision, recall, F1 score, and a confusion matrix. Regression can use metrics such as mean absolute error (MAE) and mean squared error (MSE). The metric should match the practical goal of the application.